

Cross-System Pathological Processes

Injury, inflammation, endothelial failure, ischemia, thrombosis, fibrosis, cancer, cachexia, edema, sepsis, and multiorgan dynamics

Volume 04 of the Health Atlas Research Suite

Purpose

Provide primary homes for mechanisms that traverse organ specialties, preventing repeated incomplete explanations and enabling integrated measurement and treatment.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not replace organ-specific emergency or treatment guidelines. Cross-system labels such as inflammation, dysbiosis or mitochondrial dysfunction must not be used as nonspecific diagnoses without operational evidence.

How to use this volume

Many major health problems are not organ diseases in origin or consequence. Inflammation, endothelial injury, thrombosis, ischemia, fibrosis, cancer, metabolic wasting, edema, toxicity and iatrogenesis propagate through shared blood, immune, neural, endocrine and care networks. This volume organizes those processes by mechanism, temporal phase, organ distribution and functional outcome.

Questions this volume is designed to answer

1. Which mechanisms genuinely span organs and which are only shared labels?
2. How do cross-system processes propagate and resolve?
3. Which markers reveal mechanism versus severity or consequence?
4. How should multiorgan syndromes be studied without losing causal specificity?
5. Where do supportive care, targeted therapy and rehabilitation intersect?

Scope

- Shared injury, inflammation, vascular/coagulation, fibrosis, neoplasia, metabolic and repair processes.
- Acute and chronic multisystem syndromes including sepsis, shock, cachexia and post-acute illness.
- Cross-organ imaging, biomarkers, pathology, physiology, multi-omics and monitoring.
- Cause control, support, targeted modulation, rehabilitation and system safety.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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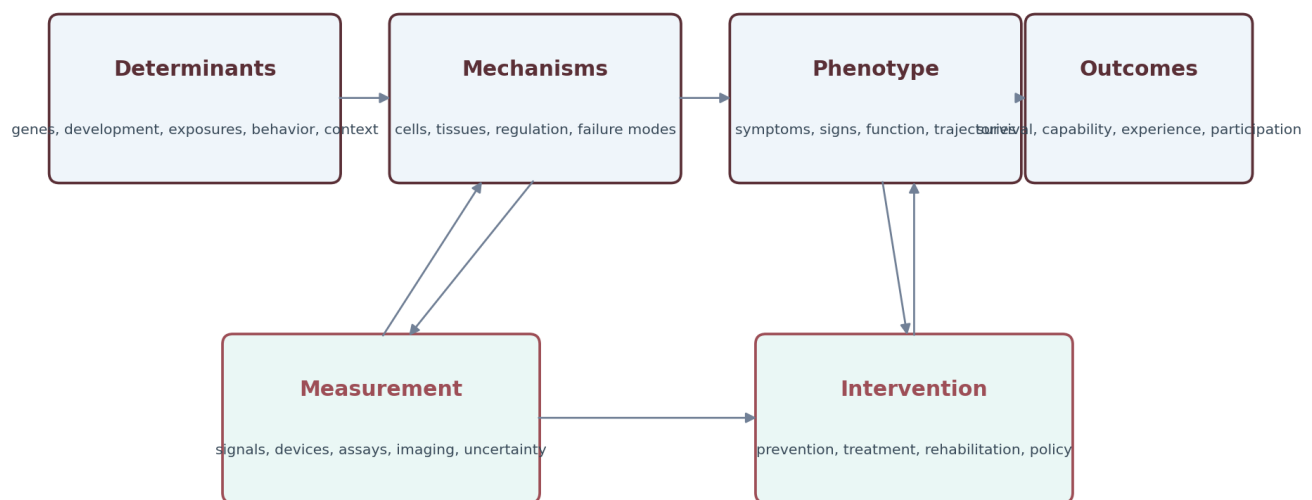
Executive synthesis

Many major health problems are not organ diseases in origin or consequence. Inflammation, endothelial injury, thrombosis, ischemia, fibrosis, cancer, metabolic wasting, edema, toxicity and iatrogenesis propagate through shared blood, immune, neural, endocrine and care networks. This volume organizes those processes by mechanism, temporal phase, organ distribution and functional outcome.

Central organizing principle

Provide primary homes for mechanisms that traverse organ specialties, preventing repeated incomplete explanations and enabling integrated measurement and treatment.

Cross-System Pathological Processes



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Shock, sepsis, stroke/ischemia, major bleeding/thrombosis, toxic exposure, rapidly progressive edema or multiorgan deterioration are time-critical. Broad mechanistic labels must not delay syndrome-specific diagnosis and cause control.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early holistic and moral models	Fever, wasting, swelling, bleeding, pain and crisis were recognized as whole-body patterns.	Preserved syndrome-level observation.	Mechanisms were humoral and nonspecific.
Anatomical and pathological organization	Cellular pathology and organ autopsy localized lesions while circulation and germ theory revealed systemic spread.	Linked syndromes to tissue injury and infection.	Organ ownership fragmented shared processes.
Statistical and laboratory medicine	Biochemistry, coagulation, imaging, intensive care and oncology quantified injury and support.	Enabled rescue and staging.	Surrogate targets and specialty-specific definitions proliferated.
Biopsychosocial and evidence-based expansion	Immunology, molecular cancer, endothelial biology and evidence-based critical care identified shared pathways and failed simplistic ones.	Improved targeted therapy and recognition of heterogeneity.	Negative trials showed mechanistic plausibility was insufficient.
Systems, data and precision era	Systems biology, multi-omics, longitudinal monitoring and rehabilitation connect acute mechanisms to chronic outcomes.	Can define dynamic endotypes and recovery.	High-dimensional associations risk relabeling rather than explanation.
Current integrated model	Current models treat cross-system pathology as interacting causal programs embedded in organ reserve, treatment and social context.	Supports explicit bridges among volumes.	Institutions still duplicate or omit ownership of multisystem problems.

2.0.1 Fever, wasting, swelling, bleeding, pain and crisis were recognized as whole-body patterns.

Scientific consequence. Preserved syndrome-level observation.

What remained unresolved. Mechanisms were humoral and nonspecific.

Evidence anchors: [3,6]

2.0.2 Cellular pathology and organ autopsy localized lesions while circulation and germ theory revealed systemic spread.

Scientific consequence. Linked syndromes to tissue injury and infection.

What remained unresolved. Organ ownership fragmented shared processes.

Evidence anchors: [3,6]

2.0.3 Biochemistry, coagulation, imaging, intensive care and oncology quantified injury and support.

Scientific consequence. Enabled rescue and staging.

What remained unresolved. Surrogate targets and specialty-specific definitions proliferated.

Evidence anchors: [3,6]

2.0.4 Immunology, molecular cancer, endothelial biology and evidence-based critical care identified shared pathways and failed simplistic ones.

Scientific consequence. Improved targeted therapy and recognition of heterogeneity.

What remained unresolved. Negative trials showed mechanistic plausibility was insufficient.

Evidence anchors: [3,6]

2.0.5 Systems biology, multi-omics, longitudinal monitoring and rehabilitation connect acute mechanisms to chronic outcomes.

Scientific consequence. Can define dynamic endotypes and recovery.

What remained unresolved. High-dimensional associations risk relabeling rather than explanation.

Evidence anchors: [3,6]

2.0.6 Current models treat cross-system pathology as interacting causal programs embedded in organ reserve, treatment and social context.

Scientific consequence. Supports explicit bridges among volumes.

What remained unresolved. Institutions still duplicate or omit ownership of multisystem problems.

Evidence anchors: [1,23,24]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Cell and tissue injury response

Core principle. Mechanical, ischemic, toxic, infectious, immune or metabolic insults trigger adaptive stress, dysfunction, death and repair.

Control architecture. Dose/time, tissue reserve, blood supply, immune response and repair capacity.

Failure modes. Reversible dysfunction, necrosis, chronic damage, scarring or organ failure.

Observable consequences. Exposure, injury biomarkers, imaging/pathology, physiology and recovery.

Cross-system coupling. All organs and biological foundations.

Evidence anchors: ^[3,4]

3.0.2 Inflammation and resolution

Core principle. Innate/adaptive mediators recruit defense and repair, then require active resolution.

Control architecture. Pattern recognition, cytokines, complement, vascular activation, efferocytosis and pro-resolving programs.

Failure modes. Acute collateral injury, chronic inflammation, immune paralysis or fibrosis.

Observable consequences. Clinical syndrome, cells/proteins, tissue and organ/function trajectory.

Cross-system coupling. Immune, infection, metabolism, coagulation and nervous systems.

Evidence anchors: ^[3,5,6]

3.0.3 Endothelial and microvascular function

Core principle. Endothelium controls barrier, tone, coagulation and cell trafficking across the circulation.

Control architecture. Shear, nitric oxide, glycocalyx, junctions, inflammatory and coagulation signals.

Failure modes. Leak, thrombosis, vasoconstriction/dilation, malperfusion and edema.

Observable consequences. Hemodynamics, permeability, vascular imaging, coagulation and organ perfusion.

Cross-system coupling. Cardiovascular, renal, lung, brain, placenta and skin.

Evidence anchors: [2,3]

3.0.4 Ischemia-reperfusion and hypoxia

Core principle. Insufficient oxygen/substrate followed by restored flow can injure cells through energy failure, ion imbalance and inflammatory/oxidative responses.

Control architecture. Flow/content/extraction, mitochondria, calcium, reactive species and immune recruitment.

Failure modes. Infarction, reperfusion injury, edema, arrhythmia or organ dysfunction.

Observable consequences. Perfusion/oxygenation, ECG/imaging, biomarkers and function.

Cross-system coupling. Cardiovascular, neurologic, renal, GI and muscle systems.

Evidence anchors: [2,3,12]

3.0.5 Coagulation, thrombosis and bleeding

Core principle. Hemostasis balances vascular/platelet/coagulation/fibrinolytic systems under flow.

Control architecture. Tissue factor, platelets, thrombin/fibrin, natural anticoagulants and endothelium.

Failure modes. Venous/arterial/microvascular thrombosis, consumption and hemorrhage.

Observable consequences. Clinical events, coagulation assays, imaging, platelets and tissue context.

Cross-system coupling. Hematology, cardiovascular, cancer, infection and pregnancy.

Evidence anchors: [2,7]

3.0.6 Fibrosis and remodeling

Core principle. Persistent injury activates stromal cells and matrix deposition with mechanical feedback.

Control architecture. TGF-beta and other signals, fibroblast states, crosslinking, inflammation and tissue strain.

Failure modes. Progressive stiffness, impaired exchange/flow and irreversible organ architecture.

Observable consequences. Imaging/elastography, organ function, biomarkers and tissue.

Cross-system coupling. Lung, liver, kidney, heart, skin, marrow and peritoneum.

Evidence anchors: [3,24]

3.0.7 Neoplasia and tumor ecology

Core principle. Clonal cells evolve under genomic, tissue, immune, metabolic and treatment selection.

Control architecture. Hallmarks of proliferation, death resistance, angiogenesis, invasion, immune escape and plasticity.

Failure modes. Local mass, metastasis, paraneoplastic/systemic effects and treatment resistance.

Observable consequences. Pathology, stage/imaging, molecular profile, function and response.

Cross-system coupling. All organs, immune, hematologic and therapeutic systems.

Evidence anchors: [8,9]

3.0.8 Metabolic dysfunction and cachexia

Core principle. Systemic energy and substrate regulation can shift toward insulin resistance, catabolism, ectopic storage or wasting.

Control architecture. Hormones, inflammation, appetite, organ disease, treatment and behavior/environment.

Failure modes. Obesity-related disease, malnutrition, sarcopenia, cachexia and impaired healing.

Observable consequences. Body composition, intake, metabolic labs, strength/function and cause.

Cross-system coupling. Endocrine, GI, muscle, cancer and social systems.

Evidence anchors: [10,11,12]

3.0.9 Edema and compartment pressure

Core principle. Fluid moves among vascular/interstitial/lymphatic spaces according to pressure, proteins, permeability and drainage.

Control architecture. Hydrostatic/oncotic forces, barrier and lymphatic transport; confined anatomy can amplify pressure.

Failure modes. Pulmonary/cerebral/peripheral edema, ascites, compartment syndrome or impaired healing.

Observable consequences. Weight/exam, imaging, pressure/perfusion, fluid chemistry and cause.

Cross-system coupling. Cardiac, renal, liver, lymphatic, brain and trauma.

Evidence anchors: [2,12,13]

3.0.10 Pain, fatigue and sickness behavior

Core principle. Cross-system symptoms coordinate protection and energy allocation but can persist beyond acute utility.

Control architecture. Nociception, cytokines, autonomic/endocrine, sleep, prediction and social context.

Failure modes. Persistent pain, fatigue, cognitive slowing, inactivity and disability.

Observable consequences. Symptom/function, organ evaluation, sleep/mood, challenge/recovery and context.

Cross-system coupling. Brain/behavior, immune, homeostasis and rehabilitation.

Evidence anchors: [14,15,16]

3.0.11 Iatrogenesis and treatment selection pressure

Core principle. Interventions change biology, ecology and behavior while creating direct and system-mediated harms.

Control architecture. Toxicity, interaction, device/procedure injury, resistance, deconditioning and diagnostic cascades.

Failure modes. Adverse drug/device events, AMR, secondary malignancy, disability or fragmented care.

Observable consequences. Treatment timeline, pharmacovigilance, incidents, function and causality assessment.

Cross-system coupling. Therapeutics, health systems, ethics and all organs.

Evidence anchors: [17,18,19]

3.0.12 Multiorgan network failure and recovery

Core principle. Organ systems depend on shared flow, chemistry, signaling and care; failure propagates and recovery is asynchronous.

Control architecture. Feedback loops, reserve, inflammatory/endocrine crosstalk, support devices and rehabilitation.

Failure modes. Shock, sepsis, critical illness, post-intensive-care syndrome and chronic multiorgan disease.

Observable consequences. Serial organ physiology/support, exposure, function/cognition and long-term trajectory.

Cross-system coupling. Integrated homeostasis, critical care, rehabilitation and whole-person outcomes.

Evidence anchors: [6,22,24]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Starling fluid exchange concept	$J_v = K_f[(P_c - P_i) - \sigma(\pi_c - \pi_i)]$	Represents hydrostatic and oncotic forces across a barrier.	Modern glycocalyx/lymphatic and nonsteady biology make simple bedside use limited.
Oxygen delivery	$DO_2 = Q C_{aO_2}$	Links blood flow and arterial oxygen content to whole-body delivery.	Regional distribution and extraction may fail despite adequate global value.
Exponential tumor growth idealization	$N(t) = N_0 e^{kt}$	A simple unconstrained clonal growth model.	Real tumors have spatial limits, immune/treatment selection and changing rates.
Attributable interaction concept	$Effect_{joint} \neq Effect_A + Effect_B$	Cross-system processes can be synergistic or antagonistic rather than additive.	The scale and causal estimand must be specified quantitatively.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within cross-system pathological processes.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within cross-system pathological processes.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: ^[4]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[10]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[17,30]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the cross-system mechanism, pathological process, multisystem syndrome, or recovery phenotype, but cross-system processes and functional effects are retained.

5.0.1 Acute inflammation

Definition and phenotype. Rapid local/systemic host response to infection, injury or immune trigger.

Pathophysiology. Vascular and leukocyte activation remove threat and initiate repair but can cause collateral damage.

Determinants and modifiers. Trigger dose/location, age, reserve, genetics, drugs and treatment timing.

Assessment. Source/trigger, local/systemic signs, targeted biomarkers/tissue and organ function.

Intervention logic. Remove trigger, supportive care and mechanism-specific modulation only when evidence supports.

Known limits and open questions. Inflammatory markers are nonspecific and suppression can impair defense.

Suite links. 03H immune; 08 infection; organ volumes.

Evidence anchors: ^[3,5]

5.0.2 Chronic low-grade inflammation

Definition and phenotype. Persistent elevation or tissue activation associated with obesity, aging, smoking, chronic infection and disease.

Pathophysiology. Metabolic, immune, barrier and tissue stress sustain signaling without acute resolution.

Determinants and modifiers. Adiposity, diet/activity opportunities, pollution, stress, sleep, infection and age.

Assessment. Cause/context, repeated markers only when actionable, organ/metabolic function and exposures.

Intervention logic. Treat causal disease/exposures and upstream determinants; avoid nonspecific “detox” claims.

Known limits and open questions. Association is broad, reverse causation common and one marker is not a diagnosis.

Suite links. 03D metabolic; 09 environment; 07 aging.

Evidence anchors: ^[10]

5.0.3 Fibrotic disease

Definition and phenotype. Progressive matrix accumulation and remodeling across one or multiple organs.

Pathophysiology. Repeated injury and fibroblast/mechanical feedback replace functional tissue with scar.

Determinants and modifiers. Autoimmune/infection, toxins, metabolic disease, drugs/radiation and genetics.

Assessment. Organ function, imaging/stiffness, biomarkers and tissue when needed.

Intervention logic. Remove driver, organ-specific antifibrotic/immune therapy and manage complications.

Known limits and open questions. Active reversible versus fixed fibrosis is hard to distinguish noninvasively.

Suite links. 03B lung; 03E liver; 03F kidney; 03A heart.

Evidence anchors: ^[3,20]

5.0.4 Thromboinflammation and microvascular injury

Definition and phenotype. Coagulation, complement, platelets and immune/endothelial activation jointly obstruct or leak microvessels.

Pathophysiology. Pathogen/damage signals and shear/barrier injury amplify clotting and inflammation.

Determinants and modifiers. Sepsis, autoimmune disease, cancer, pregnancy, devices and severe injury.

Assessment. Counts/coagulation/hemolysis, organ perfusion, imaging and mechanism-specific tests.

Intervention logic. Treat cause; anticoagulant, plasma/complement or immune therapy only by syndrome.

Known limits and open questions. Routine coagulation tests incompletely represent microvascular biology.

Suite links. 03G blood; 03H immune; 03N pregnancy.

Evidence anchors: [6,7,20]

5.0.5 Ischemia and infarction

Definition and phenotype. Insufficient tissue perfusion causes reversible dysfunction then cell death.

Pathophysiology. Flow obstruction, low pressure, embolus or microvascular failure limits oxygen/substrate.

Determinants and modifiers. Atherosclerosis/thrombosis, arrhythmia, shock, vasculitis, compression and anemia.

Assessment. Time-critical organ-specific clinical, electrical/imaging and biomarker assessment.

Intervention logic. Rapid reperfusion/source correction and organ support; secondary prevention.

Known limits and open questions. Reperfusion can injure and biomarkers lag or lack specificity.

Suite links. 03A, 03C, 03E bowel, 03F kidney.

Evidence anchors: [2,3,21]

5.0.6 Sepsis

Definition and phenotype. Life-threatening organ dysfunction due to dysregulated host response to infection.

Pathophysiology. Immune, endothelial, coagulation, autonomic, endocrine and metabolic networks lose coordinated control.

Determinants and modifiers. Source/pathogen, host reserve, AMR, access and treatment delay.

Assessment. Syndrome, cultures/molecular tests, source imaging and serial organ/perfusion measures.

Intervention logic. Prompt antimicrobials/source control and individualized support with reassessment.

Known limits and open questions. Heterogeneity limits one-target therapy and scores are not diagnostic by themselves.

Suite links. 08 infection; 03O homeostasis; 12 critical care.

Evidence anchors: [6,19]

5.0.7 Shock and reperfusion syndrome

Definition and phenotype. Circulatory failure with inadequate/maldistributed perfusion from hypovolemic, cardiogenic, obstructive or distributive causes.

Pathophysiology. Low flow/pressure, endothelial dysfunction and cellular metabolic failure interact.

Determinants and modifiers. Bleeding, infection, heart disease, embolism, anaphylaxis and iatrogenesis.

Assessment. Clinical perfusion, vitals, lactate, ultrasound/hemodynamics and cause.

Intervention logic. Cause-directed resuscitation, blood/fluid/vasoactive/mechanical support as indicated.

Known limits and open questions. Pressure normalization does not guarantee microvascular or mitochondrial recovery.

Suite links. 03O; 03A; 03G.

Evidence anchors: [2,6,17]

5.0.8 Cancer as systemic disease

Definition and phenotype. Malignancy produces local invasion/metastasis plus immune, metabolic, coagulation, endocrine and psychosocial effects.

Pathophysiology. Clonal evolution and tumor ecology interact with host/treatment.

Determinants and modifiers. Inherited/acquired drivers, exposures/infections, age and stochastic events.

Assessment. Pathology/stage/molecular classification, organ function, symptoms and goals.

Intervention logic. Prevention/detection where beneficial; surgery/radiation/systemic/immune therapy and supportive/palliative care.

Known limits and open questions. Molecular targets do not always confer benefit; screening can overdiagnose.

Suite links. 14 oncology; organ volumes; 01 whole-person.

Evidence anchors: [8,9,11]

5.0.9 Cachexia and disease-related malnutrition

Definition and phenotype. Involuntary loss of weight/muscle and function from reduced intake, inflammation/metabolism and disease/treatment.

Pathophysiology. Catabolic signaling, appetite/GI symptoms, inactivity and anabolic resistance combine.

Determinants and modifiers. Cancer, infection, organ failure, poverty and treatment toxicity.

Assessment. Weight/body composition, intake, inflammation/disease, strength/function and swallowing/GI.

Intervention logic. Treat reversible causes, symptom/nutrition support, resistance activity when feasible and goal-concordant care.

Known limits and open questions. Calories alone may not reverse cachexia; definitions and outcomes vary.

Suite links. 03E nutrition; 03I muscle; 14 palliative.

Evidence anchors: [11,16,22]

5.0.10 Edema and capillary leak

Definition and phenotype. Fluid accumulation from pressure, low oncotic force, permeability or lymphatic failure.

Pathophysiology. Barrier and Starling/lymphatic dynamics shift fluid from effective circulation to tissue.

Determinants and modifiers. Heart/kidney/liver disease, sepsis, allergy, venous/lymphatic disease and fluids.

Assessment. Distribution, weight, ultrasound/imaging, albumin/organ tests and hemodynamic context.

Intervention logic. Treat cause, mobilize fluid and protect skin/lung/brain according to compartment.

Known limits and open questions. Peripheral edema can coexist with low intravascular volume.

Suite links. 03A/F/E/H/J; 03O.

Evidence anchors: [2,12]

5.0.11 Autoimmune multisystem disease

Definition and phenotype. Loss of tolerance causes inflammation, vascular/tissue injury and organ-specific manifestations.

Pathophysiology. Autoantibody, immune-complex, cellular and innate pathways vary across patients and time.

Determinants and modifiers. Genetics, sex/hormones, infections, smoking/UV/drugs and social context.

Assessment. Clinical pattern, pretest-driven immune tests, organ assessment, tissue and damage/activity separation.

Intervention logic. Targeted immunomodulation plus organ protection, prevention and rehabilitation.

Known limits and open questions. Criteria, biomarkers and activity scores incompletely represent heterogeneous mechanisms.

Suite links. 03H immune; organ volumes.

Evidence anchors: [5,20]

5.0.12 Toxic and medication-induced multisystem injury

Definition and phenotype. A chemical or treatment damages one or several organs through dose, reactive metabolites, immune or interaction mechanisms.

Pathophysiology. Exposure, metabolism, transporters, target susceptibility and repair determine pattern.

Determinants and modifiers. Dose/timing, organ function, genetics, polypharmacy, occupation and counterfeit products.

Assessment. Exposure/treatment timeline, organ pattern, concentrations when useful and alternative causes.

Intervention logic. Stop exposure, antidote/support where indicated, report and prevent recurrence.

Known limits and open questions. Causal attribution and delayed effects are difficult; supplement products are poorly characterized.

Suite links. 09 toxicology; 14 pharmacology; 12 safety.

Evidence anchors: [17,18]

5.0.13 Post-acute and post-intensive-care syndromes

Definition and phenotype. Persistent physical, cognitive or psychological impairment after infection, trauma, surgery or critical illness.

Pathophysiology. Residual organ injury, immune/autonomic/metabolic changes, pain, sleep, deconditioning and care exposure interact.

Determinants and modifiers. Severity, baseline reserve, sedation/immobility, social support and access to follow-up.

Assessment. Multidomain symptoms/function/cognition/mental health plus targeted organ/challenge testing.

Intervention logic. Coordinated rehabilitation, symptom/cause treatment and social/vocational support.

Known limits and open questions. Broad labels mix mechanisms and trials are heterogeneous.

Suite links. 01 whole-person; 10 rehab; 05/06; organ volumes.

Evidence anchors: ^[16,22,23]

5.0.14 Multiorgan failure and chronic organ crosstalk

Definition and phenotype. Failure in one organ propagates through flow, chemistry, inflammation and treatment to others.

Pathophysiology. Cardiorenal, hepatorenal, brain-body and other feedback loops create coupled trajectories.

Determinants and modifiers. Sepsis, shock, chronic disease, age, treatment and fragmented care.

Assessment. Serial organ measures, causal/treatment timeline, function and competing risks.

Intervention logic. Prioritize reversible drivers and total benefit; coordinate support and long-term recovery/palliative goals.

Known limits and open questions. Attribution to one “primary” organ can be arbitrary and scores miss interactions.

Suite links. 03O homeostasis; 12 systems; 01 outcomes.

Evidence anchors: ^[6,24]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current cross-system pathological processes risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current cross-system pathological processes risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

Evidence anchors: ^[28]

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[16,23]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[3]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

Evidence anchors: ^[28]

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[23]

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Evidence anchors: ^[26]

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Multisystem history and timeline	Noninvasive	Symptoms, exposures, infections, drugs/procedures and organ events aligned in time.	Causal candidates and interaction map.	Often reveals cross-system sequence.	Recall and fragmented records.
Serial vital signs and organ panels	Noninvasive/minimally invasive	Vital signs, renal, liver, hematologic, metabolic and inflammatory labs.	Deterioration, organ pattern and response.	Scalable longitudinal surveillance.	Nonspecific markers, treatment effects and threshold alert burden.
Point-of-care ultrasound	Noninvasive	Focused cardiac, lung, vascular, abdominal and soft-tissue ultrasound.	Flow, fluid, obstruction, organ morphology and procedural support.	Rapid cross-organ bedside context.	Operator, limited windows and overconfidence.
Whole-body CT/MRI/PET	Noninvasive to contrast/radiotracer	Cross-sectional anatomy, diffusion/perfusion and molecular tracer uptake.	Injury, infection source, tumor, inflammation and organ burden.	Comprehensive spatial survey.	Radiation/contrast, incidental findings, cost and nonspecific uptake.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Inflammatory and endothelial biomarkers	Minimally invasive sample	CRP/ESR, cytokines, complement, adhesion/barrier and vascular injury markers.	Inflammatory burden and candidate endotype.	Accessible trend for selected syndromes.	Low specificity, timing and assay/batch effects.
Coagulation and microangiopathy panel	Minimally invasive sample	Counts, PT/aPTT, fibrinogen/D-dimer, hemolysis and specialized assays.	Consumption, thrombosis, bleeding and mechanism clues.	Actionable in defined syndromes.	Routine tests miss endothelium/flow and are drug-sensitive.
Organ-injury biomarkers	Minimally invasive sample	Troponin, creatinine, transaminases, neuronal and other proteins/metabolites.	Cell injury or organ function proxy.	Sensitive and scalable.	Source specificity, chronic elevation, kinetics and surrogate overuse.
Tissue biopsy and integrated pathology	Invasive	Histology, microbiology, immune deposits, fibrosis and molecular pathology.	Direct pattern and etiology support.	Can separate processes with similar systemic markers.	Sampling, procedural risk and treatment timing.
Microcirculation and tissue oxygen sensing	Noninvasive/minimally invasive	NIRS, capillary/laser flow, tissue oxygen and microscopy.	Regional perfusion/oxygenation proxies.	Potential view beyond global pressure.	Depth/site/scattering and uncertain thresholds.
Body composition and nutritional assessment	Noninvasive/imaging	Weighting trajectory, DXA/CT, bioimpedance, strength and dietary assessment.	Fat/muscle/fluid and malnutrition/cachexia risk.	Connects systemic disease to reserve.	Edema, device models and self-report.
Composite organ-dysfunction scores	Software	Weighted vitals, labs and support interventions.	Severity, prognosis and trial stratification.	Standardizes population comparison.	Treatment circularity, missingness, not causal or individual destiny.
Multi-omics and systems phenotyping	Sample/software research	Genomic, transcript, protein, metabolite, microbiome and clinical integration.	Candidate endotypes and pathways.	May resolve heterogeneous syndromes.	Batch, confounding, overfitting and unstable clusters.
Wearable deterioration and recovery sensing	Noninvasive	Heart/respiratory rate, temperature, activity, sleep and symptoms.	Within-person change and recovery trajectory.	Continuous low-burden context.	Nonspecific signals, missingness and alert fatigue.

7.0.1 Multisystem history and timeline

Measured quantity and transducer. Symptoms, exposures, infections, drugs/procedures and organ events aligned in time.

Typical outputs. Causal candidates and interaction map.

Strengths. Often reveals cross-system sequence.

Limitations and artifacts. Recall and fragmented records.

Reference and validation strategy. Source reconciliation and prospective follow-up.

Clinical/research maturity. Established.

Evidence anchors: ^[17]

7.0.2 Serial vital signs and organ panels

Measured quantity and transducer. Vitals plus renal, liver, hematologic, metabolic and inflammatory labs.

Typical outputs. Deterioration, organ pattern and response.

Strengths. Scalable longitudinal surveillance.

Limitations and artifacts. Nonspecific markers, treatment effects and threshold alert burden.

Reference and validation strategy. Device/lab QA and outcome-linked trend rules.

Clinical/research maturity. Established.

Evidence anchors: ^[6]

7.0.3 Point-of-care ultrasound

Measured quantity and transducer. Focused cardiac, lung, vascular, abdominal and soft-tissue ultrasound.

Typical outputs. Flow, fluid, obstruction, organ morphology and procedural support.

Strengths. Rapid cross-organ bedside context.

Limitations and artifacts. Operator, limited windows and overconfidence.

Reference and validation strategy. Training, image QA, expert/reference and clinical utility.

Clinical/research maturity. Established selected.

Evidence anchors: ^[2]

7.0.4 Whole-body CT/MRI/PET

Measured quantity and transducer. Cross-sectional anatomy, diffusion/perfusion and molecular tracer uptake.

Typical outputs. Injury, infection source, tumor, inflammation and organ burden.

Strengths. Comprehensive spatial survey.

Limitations and artifacts. Radiation/contrast, incidental findings, cost and nonspecific uptake.

Reference and validation strategy. Protocol, pathology/outcome and pretest-driven use.

Clinical/research maturity. Established selected.

Evidence anchors: [9,17]

7.0.5 Inflammatory and endothelial biomarkers

Measured quantity and transducer. CRP/ESR, cytokines, complement, adhesion/barrier and vascular injury markers.

Typical outputs. Inflammatory burden and candidate endotype.

Strengths. Accessible trend for selected syndromes.

Limitations and artifacts. Low specificity, timing and assay/batch effects.

Reference and validation strategy. Clinical/tissue correlation and treatment-interaction validation.

Clinical/research maturity. Established narrow; research broad.

Evidence anchors: [5,25]

7.0.6 Coagulation and microangiopathy panel

Measured quantity and transducer. Counts, PT/aPTT, fibrinogen/D-dimer, hemolysis and specialized assays.

Typical outputs. Consumption, thrombosis, bleeding and mechanism clues.

Strengths. Actionable in defined syndromes.

Limitations and artifacts. Routine tests miss endothelium/flow and are drug-sensitive.

Reference and validation strategy. Standard preanalytics and clinical-event validation.

Clinical/research maturity. Established.

Evidence anchors: [7]

7.0.7 Organ-injury biomarkers

Measured quantity and transducer. Troponin, creatinine, transaminases, neuronal and other proteins/metabolites.

Typical outputs. Cell injury or organ function proxy.

Strengths. Sensitive and scalable.

Limitations and artifacts. Source specificity, chronic elevation, kinetics and surrogate overuse.

Reference and validation strategy. Method-specific reference/change and organ outcome.

Clinical/research maturity. Established selected.

Evidence anchors: [3,28]

7.0.8 Tissue biopsy and integrated pathology

Measured quantity and transducer. Histology, microbiology, immune deposits, fibrosis and molecular pathology.

Typical outputs. Direct pattern and etiology support.

Strengths. Can separate processes with similar systemic markers.

Limitations and artifacts. Sampling, procedural risk and treatment timing.

Reference and validation strategy. Adequate site and clinicopathologic integration.

Clinical/research maturity. Established.

Evidence anchors: [3,9]

7.0.9 Microcirculation and tissue oxygen sensing

Measured quantity and transducer. NIRS, capillary/laser flow, tissue oxygen and microscopy.

Typical outputs. Regional perfusion/oxygenation proxies.

Strengths. Potential view beyond global pressure.

Limitations and artifacts. Depth/site/scattering and uncertain thresholds.

Reference and validation strategy. Phantoms, invasive comparator and outcome utility.

Clinical/research maturity. Research/selected.

Evidence anchors: ^[2]

7.0.10 Body composition and nutritional assessment

Measured quantity and transducer. Weight trajectory, DXA/CT, bioimpedance, strength and dietary assessment.

Typical outputs. Fat/muscle/fluid and malnutrition/cachexia risk.

Strengths. Connects systemic disease to reserve.

Limitations and artifacts. Edema, device models and self-report.

Reference and validation strategy. Reference imaging/function and longitudinal outcomes.

Clinical/research maturity. Established selected.

Evidence anchors: ^[10,22]

7.0.11 Composite organ-dysfunction scores

Measured quantity and transducer. Weighted vitals, labs and support interventions.

Typical outputs. Severity, prognosis and trial stratification.

Strengths. Standardizes population comparison.

Limitations and artifacts. Treatment circularity, missingness, not causal or individual destiny.

Reference and validation strategy. External calibration and decision-impact restriction.

Clinical/research maturity. Established.

Evidence anchors: ^[6,27]

7.0.12 Multi-omics and systems phenotyping

Measured quantity and transducer. Genomic, transcript, protein, metabolite, microbiome and clinical integration.

Typical outputs. Candidate endotypes and pathways.

Strengths. May resolve heterogeneous syndromes.

Limitations and artifacts. Batch, confounding, overfitting and unstable clusters.

Reference and validation strategy. Locked multisite validation and treatment-interaction trials.

Clinical/research maturity. Frontier.

Evidence anchors: ^[25,26]

7.0.13 Wearable deterioration and recovery sensing

Measured quantity and transducer. Heart/respiratory rate, temperature, activity, sleep and symptoms.

Typical outputs. Within-person change and recovery trajectory.

Strengths. Continuous low-burden context.

Limitations and artifacts. Nonspecific signals, missingness and alert fatigue.

Reference and validation strategy. Reference events and randomized response pathways.

Clinical/research maturity. Emerging.

Evidence anchors: ^[27]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise cross-system pathological processes quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise cross-system pathological processes quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

Evidence anchors: ^[26]

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[26]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: ^[27]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[27]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Cause/source control	Acute/causal	Remove infection source, restore blood flow, stop toxin/drug, control bleeding or relieve obstruction.	Often the decisive intervention across cross-system syndromes.	Procedure delay/harm and incomplete source identification.	Source resolution, physiology and organ outcome.
Protective organ support	Acute/supportive	Support oxygenation, circulation, renal/metabolic and nutrition while minimizing device/treatment injury.	Life-saving while cause resolves.	Ventilator, fluid, catheter, sedation and immobility harms.	Organ function, support intensity/duration and complications.
Anti-inflammatory and immunomodulatory therapy	Treatment	Suppress or redirect harmful immune pathways in defined syndromes.	Strong in selected autoimmune/inflammatory disease; many broad sepsis approaches fail.	Infection, malignancy and organ/metabolic toxicity.	Disease activity, organ function, infection and long-term damage.
Antithrombotic and hemostatic treatment	Treatment	Prevent/lyse clot or restore platelets/factors and control bleeding.	High benefit when mechanism and absolute risk are correct.	Bleeding/thrombosis and monitoring complexity.	Clinical events, labs where relevant and organ response.
Antifibrotic and remodeling intervention	Disease modification	Remove ongoing injury and inhibit selected profibrotic pathways.	Established for narrow organ diseases; broad reversal is limited.	Organ/drug toxicity and treatment of fixed scar without benefit.	Organ function, imaging/stiffness, symptoms and survival.
Cancer prevention and multimodal therapy	Prevention/treatment	Remove carcinogens/infections and combine local/systemic/immune approaches by stage/biology.	Major survival benefit for many defined cancers.	Toxicity, resistance, overdiagnosis and inequity.	Survival, response, function, toxicity and goals.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Nutrition and anabolic rehabilitation	Supportive/rehabilitation	Address barriers and stimulate muscle/function with nutrition and activity where feasible.	Useful for malnutrition/deconditioning; cachexia response varies.	Refeeding, aspiration and burden.	Intake, weight/composition, strength and function.
Symptom, pain and fatigue management	Supportive	Treat causal and central/peripheral symptom mechanisms while preserving activity.	Improves quality and participation when integrated.	Sedation, polypharmacy and invalidation.	Symptoms, function, sleep and adverse effects.
Prehabilitation and complication prevention	Prevention	Improve reserve and reduce infection, thrombosis, delirium and deconditioning around major treatment.	Promising/established in selected surgery/cancer pathways.	Selection, delay and unequal access.	Complications, function, length of stay and recovery.
Medication/device safety and de-implementation	System	Stop low-value harmful interventions and redesign processes.	Can reduce adverse events and resistance.	Withdrawal/undertreatment and institutional resistance.	Adverse events, total burden and outcomes.
Integrated recovery and palliative care	Whole-person/system	Coordinate organ follow-up, rehabilitation, mental health, social and goal-concordant support.	Addresses long-term burden after complex illness.	Fragmentation and access.	Function, readmission, symptoms, goals and caregiver outcomes.

9.0.1 Cause/source control

Target and mechanism. Remove infection source, restore blood flow, stop toxin/drug, control bleeding or relieve obstruction.

Evidence and expected outcome. Often the decisive intervention across cross-system syndromes.

Risks, burden, and equity. Procedure delay/harm and incomplete source identification.

Response monitoring. Source resolution, physiology and organ outcome.

Evidence anchors: [6,17]

9.0.2 Protective organ support

Target and mechanism. Support oxygenation, circulation, renal/metabolic and nutrition while minimizing device/treatment injury.

Evidence and expected outcome. Life-saving while cause resolves.

Risks, burden, and equity. Ventilator, fluid, catheter, sedation and immobility harms.

Response monitoring. Organ function, support intensity/duration and complications.

Evidence anchors: [17,22]

9.0.3 Anti-inflammatory and immunomodulatory therapy

Target and mechanism. Suppress or redirect harmful immune pathways in defined syndromes.

Evidence and expected outcome. Strong in selected autoimmune/inflammatory disease; many broad sepsis approaches fail.

Risks, burden, and equity. Infection, malignancy and organ/metabolic toxicity.

Response monitoring. Disease activity, organ function, infection and long-term damage.

Evidence anchors: [6,20]

9.0.4 Antithrombotic and hemostatic treatment

Target and mechanism. Prevent/lyse clot or restore platelets/factors and control bleeding.

Evidence and expected outcome. High benefit when mechanism and absolute risk are correct.

Risks, burden, and equity. Bleeding/thrombosis and monitoring complexity.

Response monitoring. Clinical events, labs where relevant and organ response.

Evidence anchors: [2,7]

9.0.5 Antifibrotic and remodeling intervention

Target and mechanism. Remove ongoing injury and inhibit selected profibrotic pathways.

Evidence and expected outcome. Established for narrow organ diseases; broad reversal is limited.

Risks, burden, and equity. Organ/drug toxicity and treatment of fixed scar without benefit.

Response monitoring. Organ function, imaging/stiffness, symptoms and survival.

Evidence anchors: [3,28]

9.0.6 Cancer prevention and multimodal therapy

Target and mechanism. Reduce carcinogens/infections and combine local/systemic/immune approaches by stage/biology.

Evidence and expected outcome. Major survival benefit for many defined cancers.

Risks, burden, and equity. Toxicity, resistance, overdiagnosis and inequity.

Response monitoring. Survival, response, function, toxicity and goals.

Evidence anchors: [8,9]

9.0.7 Nutrition and anabolic rehabilitation

Target and mechanism. Address intake barriers and stimulate muscle/function with nutrition and activity where feasible.

Evidence and expected outcome. Useful for malnutrition/deconditioning; cachexia response varies.

Risks, burden, and equity. Refeeding, aspiration and burden.

Response monitoring. Intake, weight/composition, strength and function.

Evidence anchors: [11,22]

9.0.8 Symptom, pain and fatigue management

Target and mechanism. Treat causal and central/peripheral symptom mechanisms while preserving activity.

Evidence and expected outcome. Improves quality and participation when integrated.

Risks, burden, and equity. Sedation, polypharmacy and invalidation.

Response monitoring. Symptoms, function, sleep and adverse effects.

Evidence anchors: [14,16]

9.0.9 Prehabilitation and complication prevention

Target and mechanism. Improve reserve and reduce infection, thrombosis, delirium and deconditioning around major treatment.

Evidence and expected outcome. Promising/established in selected surgery/cancer pathways.

Risks, burden, and equity. Selection, delay and unequal access.

Response monitoring. Complications, function, length of stay and recovery.

Evidence anchors: [22,29]

9.0.10 Medication/device safety and de-implementation

Target and mechanism. Stop low-value harmful interventions and redesign processes.

Evidence and expected outcome. Can reduce adverse events and resistance.

Risks, burden, and equity. Withdrawal/undertreatment and institutional resistance.

Response monitoring. Adverse events, total burden and outcomes.

Evidence anchors: ^[17,30]

9.0.11 Integrated recovery and palliative care

Target and mechanism. Coordinate organ follow-up, rehabilitation, mental health, social and goal-concordant support.

Evidence and expected outcome. Addresses long-term burden after complex illness.

Risks, burden, and equity. Fragmentation and access.

Response monitoring. Function, readmission, symptoms, goals and caregiver outcomes.

Evidence anchors: ^[11,22]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can sepsis and other syndromes be endotyped for targeted therapy?	Biological heterogeneity and reproducible clusters exist.	Stable causal subgroups with prospective treatment interaction.	Timing, treatment and platform drive cluster assignment.	Rapid locked assays embedded in randomized adaptive trials.
Can fibrosis be reversed across organs?	Some early remodeling is modifiable and selected antifibrotics slow progression.	Mechanisms and windows for substantial functional reversal.	Matrix, cell state and organ architecture differ; biomarkers poorly distinguish activity.	Mechanism-specific longitudinal tissue/imaging and combination trials.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within cross-system pathological processes.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can sepsis and other syndromes be endotyped for targeted therapy?

Established knowledge. Biological heterogeneity and reproducible clusters exist.

Unresolved issue. Stable causal subgroups with prospective treatment interaction.

Why current evidence is insufficient. Timing, treatment and platform drive cluster assignment.

Research that would change confidence. Rapid locked assays embedded in randomized adaptive trials.

Evidence anchors: [6,25]

10.0.2 Can fibrosis be reversed across organs?

Established knowledge. Some early remodeling is modifiable and selected antifibrotics slow progression.

Unresolved issue. Mechanisms and windows for substantial functional reversal.

Why current evidence is insufficient. Matrix, cell state and organ architecture differ; biomarkers poorly distinguish activity.

Research that would change confidence. Mechanism-specific longitudinal tissue/imaging and combination trials.

Evidence anchors: [3,24]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within cross-system pathological processes.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: [16,27]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[9,28]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[16,23]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Cross-organ acute-illness atlas	Which dynamic pathways predict recovery versus persistent multiorgan dysfunction?	Prospective sepsis/trauma/surgery cohort with standardized serial omics, physiology and one-year function.	Replicated causal mediators and treatment-response hypotheses.	Static severity signatures without incremental or actionable value.
0. Mechanism-to-function trial standard	Can targeted process trials require both target engagement and patient-important benefit?	Platform protocol with preregistered mechanism, organ, function and harm endpoints.	Interventions advance only when mechanistic and clinical criteria align.	Surrogate improvement drives adoption despite no lived benefit.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for cross-system pathological processes?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Cross-organ acute-illness atlas

Question. Which dynamic pathways predict recovery versus persistent multiorgan dysfunction?

Design. Prospective sepsis/trauma/surgery cohort with standardized serial omics, physiology and one-year function.

Success criterion. Replicated causal mediators and treatment-response hypotheses.

Failure or stop criterion. Static severity signatures without incremental or actionable value.

Evidence anchors: [22,26]

11.0.2 0. Mechanism-to-function trial standard

Question. Can targeted process trials require both target engagement and patient-important benefit?

Design. Platform protocol with preregistered mechanism, organ, function and harm endpoints.

Success criterion. Interventions advance only when mechanistic and clinical criteria align.

Failure or stop criterion. Surrogate improvement drives adoption despite no lived benefit.

Evidence anchors: [28]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for cross-system pathological processes?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: ^[26]

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[27]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: [28,29]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: [23]

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: [29]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
02	Biological Foundations	Cell injury, signaling, metabolism, matrix and clonal evolution provide lower-level mechanisms.	a process requires molecular/cellular detail.
030	Integrated Homeostatic Systems	Shock, oxygen delivery, chemistry, thermal/metabolic and recovery dynamics.	system stability or reserve is failing.
08	Infection, Immunity, Host-Microbe and Biosecurity	Agents, host response and AMR contextualize sepsis and chronic infection.	infection or outbreak is causal.
10	Function, Disability, Rehabilitation and Participation	Post-acute disability, cachexia, pain and recovery require functional outcomes.	survival is not the endpoint.
14	Interventions and Therapeutics	Shared pharmacology, procedures, devices and supportive care determine cross-system benefit and harm.	targeting a mechanism or supporting organs.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Resolution	Active termination and cleanup phase of inflammation that restores or adapts tissue homeostasis.
Fibrosis	Pathologic or excessive extracellular-matrix deposition and remodeling that impairs tissue architecture/function.
Endothelial dysfunction	Impaired vascular barrier, tone, antithrombotic or signaling functions; requires operational definition.
Multiorgan dysfunction	Concurrent or sequential impairment of multiple organ systems, often through shared and propagating mechanisms.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.